

Nilay Tripathi

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🔗 [nilayt321.github.io](https://github.com/nilayt321)

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EDUCATION:

Michigan State University East Lansing, MI 09/25–Present
Doctor of Philosophy, Student

- Second year doctoral student
- Intended area of study: low dimensional topology

Rutgers University–New Brunswick New Brunswick, NJ 09/21–05/25
Undergraduate Degree

- Summa Cum Laude, CGPA: 3.907
- B.S. in Mathematics, High Honors
- B.A. in Statistics, Highest Honors
- B.A. in Computer Science

AREAS OF INTEREST:

Low dimensional topology, including knot theory, quantum topology, manifolds

TALKS & PRESENTATIONS:

At Rutgers University:

- “Integrable Systems in Hamiltonian Dynamics” 05/25
Independent Study Talk
- “Homology & Its Applications” 11/24
Undergraduate Seminar Talks by RUMA
- “Smooth Manifolds & Symplectic Geometry” 09/24
Directed Reading Program
- “de Rham Cohomology” 05/24
Part of Math 412 at Rutgers
- “Banach & Hilbert Spaces: An Introduction to Functional Analysis” 12/23
Part of Math 441 at Rutgers
- “An Overview of Hermite Polynomials” 12/23
Directed Reading Program

MENTORSHIP EXPERIENCE:

- **Summer Topology Program at MSU** 05/26–06/26
Program mentor for undergraduates to learn about a topic adjacent to topology/geometry.

TEACHING EXPERIENCE:

At Michigan State University:

As Teaching Assistant (TA):

- MTH 124: Survey of Calculus I 06/26–08/26

Mathematics Learning Center Tutor:

- MTH 132: Calculus I 01/26–05/26
- MTH 234: Multivariable Calculus 09/25–12/25

At Rutgers University:

Undergraduate Learning Assistant:

- CS 112: Data Structures 01/25–05/25
- CS 111: Introductory Computer Science 09/24–12/24
- Math 300: Introduction to Mathematical Reasoning 09/23–05/24
- Stat 212: Statistics II 09/22–05/23

Math Department Grader:

- Math 412: Mathematical Analysis II 01/25–05/25
- Math 350: Linear Algebra 01/25–05/25
- Math 441: Introductory Topology I 09/24–12/24
- Math 311: Introductory Real Analysis I 09/24–12/24

RELEVANT COURSEWORK:

At MSU (Graduate):

- MTH 868, Differential Topology Spring 2026
- MTH 828/829, Real/Complex Analysis Fall 2025, Spring 2026
- MTH 818/819, Graduate Algebra I/II Fall 2025, Spring 2026

At Rutgers (Undergraduate):

- Math 540/541, Algebraic Topology I/II (Graduate) Fall 2024, Spring 2025
- Math 451/452, Abstract Algebra I/II (Honors) Fall 2024, Spring 2025
- Math 492, Honors Junior-Senior Seminar Spring 2025
- Math 503, Complex Analysis I Fall 2024
- Math 441/442, Introductory Topology I/II Fall 2023, Spring 2024
- Math 411/412, Mathematical Analysis I/II (Honors) Fall 2023, Spring 2024
- Math 491, Problem-Solving Math Seminar Fall 2023
- Math 350, Linear Algebra (Honors) Spring 2023

HONORS & AWARDS:

At Rutgers (Undergraduate):

- Summa Cum Laude
- Dean's List
- Student presentation award for my DRP presentation on Hermite polynomials, given in Fall 2023

SKILLS:

- Languages: English (native), Hindi (native), French (beginner)
- Programming Languages: Python, C/C++, Java, R, Matlab, $\text{\LaTeX}$
- Software Skills: Git/GitHub, MS Office, Linux (Ubuntu, Fedora/RHEL), Jupyter Notebook/JupyterLab